

Guangzhi Chen

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EDUCATION

The University of Texas at Austin

August 2024 - May 2027

Bachelor of Science, Aerospace Engineering

Cumulative GPA: 3.91/4.00

Relevant Courses: Linear System Analysis, Feedback Control System, Spacecraft Dynamics

EXPERIENCE

GNC Research Intern, **CDUS Lab (Dr. Maruthi Akella) - USSF SSTI**

June 2026 - July 2026

- Rebuilt a pose-estimation Multiplicative EKF around a 15-state Relative Orbital Element translation model, replacing a gravity-blind random walk with an exact Keplerian state-transition matrix
- Derived the ROE-to-camera measurement map and its 3x6 Jacobian, resolving the 3-measurement/6-state rank deficiency with a diffuse-prior initialization seeding the minimum-norm solution
- Validated on Stanford SLAB's SHIRT dataset (2,371 frames, 2 orbits of J2-perturbed truth): cut position RMSE from 0.132 m to 0.020 m, a 7x gain over baseline and below the 0.138 m measurement noise floor
- Derived a numeric ROE control-input matrix to inject known chaser impulse burns into the predict step of EKF, cutting maneuver-node innovation from 2.12 m to 0.047 m

Software Engineer, **Design Build Vertical Flight**

August 2025 - Present

- Interfaced with ArduPilot in Python to send high-level waypoint commands in a Gazebo simulation, designing an algorithm to convert metric distances into GPS coordinates for navigation testing.
- Configured the flight controller (Pixhawk 6x mini) and integrated core avionics subsystems, including a Raspberry Pi 4, GPS module, and radio telemetry.

Software Engineer, **Robotics Automation Society - RoboMaster**

September 2025 - Present

- Transitioned to the software team to develop sensor and vision systems; integrated LiDAR for rear obstacle detection and implemented an OpenCV computer vision pipeline to track opponent robots by filtering live video frames for bright red/blue light signatures, and choosing targets

Activities

Controls Engineer, **Quadcopter Controls Simulation (Personal Project)**

August 2025 - May 2026

- Developed a 6-DOF Quadcopter Simulator in C++, utilizing Eigen3 for rigid-body dynamics and implementing Runge-Kutta 4 (RK4)
- Implemented a safety-first motor mixing strategy that resolves actuator saturation by prioritizing thrust over attitude commands, preventing fly-aways and loss of altitude control.
- Built a thrust-vector flight controller using cross-product-based attitude error rather than Euler angles, avoiding gimbal lock and enabling stable high-angle flight through a PID controller

Officer, **LUNAR Undergraduate Council, UT Aerospace**

November 2024 - Present

- Re-established the department's undergraduate advisory council post-COVID and launched the inaugural Student-Faculty Research Mixer, coordinating faculty to connect undergraduates with research opportunities.
- Helped launch an alumni mentorship program (integrated with Cockrell's CAMP), recruiting 10+ senior aerospace/defense alumni — including NASA and Lockheed Orion GNC engineers — as mentors.

SKILLS & ACCOMPLISHMENTS

- **Technical Skills:** Python, C++, ROS 2, Gazebo, OpenCV, OOP, GitHub, SolidWorks, FEA, Soldering
- **Languages:** Chinese - fluent
- **Volunteering:** GivePulse, Storehouse of Collin County, Kids Against Hunger, ABC English Learning